Sequences & Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Arithmetic sequences

For the arithmetic sequence with $a_1 = 6$ and common difference $d = 4$, find a_{12}.

$a_{12} = a_1 + (12 - 1)d = 6 + 11(4) = 50.$

2 [Foundational] • Target: Arithmetic sequences

For the arithmetic sequence with $a_1 = 20$ and $d = -3$, find a_{15}.

$a_{15} = 20 + 14(-3) = 20 - 42 = -22.$

3 [Foundational] • Target: Arithmetic sequences

A geometric sequence has $a_1 = 3$ and each term is 2 times the previous one. Find the smallest n for which $a_n = 384$.

$a_n = 3 \cdot 2^{n-1}$. Setting $3 \cdot 2^{n-1} = 384$ gives $2^{n-1} = 128 = 2^7$, so $n - 1 = 7$ and $n = 8$.

4 [Foundational] • Target: Arithmetic series (sum formula)

Find S_{12} for the arithmetic sequence with $a_1 = 6$, $d = 4$ (using $a_{12} = 50$ from above).

$S_{12} = \frac{n(a_1 + a_n)}{2} = \frac{12(6 + 50)}{2} = 6(56) = 336.$

5 [Foundational] • Target: Arithmetic series (sum formula)

Find S_8 for the arithmetic sequence with $a_1 = 2$, $d = 5$.

$a_8 = 2 + 7(5) = 37$. $S_8 = \frac{8(2 + 37)}{2} = 4(39) = 156.$

6 [Foundational] • Target: Geometric sequences

For the geometric sequence with $a_1 = 4$ and common ratio $r = 3$, find a_6.

$a_6 = a_1 r^5 = 4(3^5) = 4(243) = 972.$

7 [Foundational] • Target: Geometric sequences

A geometric sequence begins $5, -10, 20, -40, \ldots$. Find r and a_7.

$r = \frac{-10}{5} = -2$. $a_7 = 5(-2)^6 = 5(64) = 320.$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Geometric series (finite)

Find S_6 for the geometric sequence with $a_1 = 4$, $r = 3$ (using $a_6 = 972$ from above).

$S_6 = \frac{a_1(r^6 - 1)}{r - 1} = \frac{4(729 - 1)}{2} = \frac{4(728)}{2} = 1456.$

9 [Intermediate] • Target: Geometric series (finite)

Find S_5 for the geometric sequence $5, -10, 20, -40, \ldots$.

$S_5 = \frac{5|(-2)^5 - 1|}{-2 - 1} = \frac{5(-32 - 1)}{-3} = \frac{5(-33)}{-3} = 55.$

10 [Intermediate] • Target: Infinite geometric series

Find the sum of the infinite geometric series $20 + \frac{20}{3} + \frac{20}{9} + \cdots$.

$a = 20$, $r = \frac{1}{3}$, and $|r| < 1$, so by Theorem 3.2 the sum is $\frac{20}{1 - \frac{1}{3}} = \frac{20}{\frac{2}{3}} = 30$.

11 [Intermediate] • Target: Infinite geometric series

Explain why the infinite geometric series with $a = 5$, $r = 2$ has no finite sum.

Theorem 3.2 requires $|r| < 1$ for the partial sums to converge. Here $|r| = 2 > 1$, so the terms grow without bound and the series diverges — it has no finite sum.

12 [Intermediate] • Target: Arithmetic series (sum formula)

Express the repeating decimal $0.3\overline{18} = 0.3181818\ldots$ as an irreducible fraction using the infinite geometric series formula.

Step 1: Separate the non-repeating and repeating components: $0.3\overline{18} = \frac{3}{10} + 0.0181818\ldots$ Step 2: Represent the repeating part as an infinite geometric series: $0.0181818\ldots = \frac{18}{1000} + \frac{18}{100000} + \cdots$ Here the initial term is $a_1 = \frac{18}{1000}$ and the common ratio is $r = \frac{1}{100}$. Step 3: Sum the geometric series.

13 [Intermediate] • Target: Geometric sequences

The second term of a geometric sequence is 12 and the fifth term is $\frac{32}{9}$. Find the common ratio r, the first term a_1, and the sum of the infinite series S_∞.

Step 1: Express terms in terms of a_1 and r: $a_2 = a_1 r = 12$ $a_5 = a_1 r^4 = \frac{32}{9}$ Step 2: Divide a_5 by a_2: $\frac{a_1 r^4}{a_1 r} = r^3 = \frac{32/9}{12} = \frac{32}{108} = \frac{8}{27}$ Step 3: Extract the common ratio r: $r = \sqrt[3]{\frac{8}{27}} = \mathbf{\frac{2}{3}}$. Step 4: Find a_1: $a_1\left(\frac{2}{3}\right) = 12 \Rightarrow a_1 = 12 \times \frac{3}{2} = \mathbf{18}$. Step 5: Compute the infinite sum S_∞: Since \(|r| = \frac{2}{3}

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14 **[Intermediate]** • **Target: Geometric series (finite)**

In an arithmetic sequence, the 4th term is 17 and the 10th term is 41. Determine the common difference d, the first term a_1, and the partial sum of terms from a_{11} through a_{25} inclusive.

Step 1: Solve for common difference d: $a_{10} - a_4 = (10-4)d = 6d$ $41 - 17 = 24 \Rightarrow 6d = 24 \Rightarrow \mathbf{d = 4}$. Step 2: Solve for the first term a_1: $a_4 = a_1 + 3d \Rightarrow 17 = a_1 + 3(4) \Rightarrow a_1 = 17 - 12 = \mathbf{5}$. Step 3: Find boundary terms for the range $n = 11$ to 25: Number of terms: $N = 25 - 11 + 1 = 15$. Term $a_{11} = a_1 + 10d = 5 + 10(4) = 45$. Term $a_{25} = a_1 + 24d = 5 + 24(4) = 5 + 96 = 101$. Step 4: Compute the sum.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Infinite geometric series** • **Cross-Lesson: 3.01 Matrices & Vectors**

Evaluate the exact value of the telescoping sum: $\sum_{k=1}^{50} \left(\frac{1}{k} - \frac{1}{k+1}\right)$ and evaluate the convergent infinite series: $\sum_{n=1}^{\infty} \frac{3}{4^n}$.

Part 1: Telescoping sum: Write out the first few and last terms: $\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{50} - \frac{1}{51}\right)$ All intermediate terms cancel pairwise, leaving: $1 - \frac{1}{51} = \mathbf{\frac{50}{51}}$. Part 2: Infinite series: Factor the numerator and identify initial term and ratio: $\sum_{n=1}^{\infty} \frac{3}{4^n} = \frac{3}{4} + \frac{3}{16} + \frac{3}{64} + \cdots$ Initial term $a_1 = \frac{3}{4}$, ratio $r = \frac{1}{4}$..

16 **[Challenging]** • **Target: Arithmetic sequences** • **Cross-Lesson: 4.01 Advanced Functions & Equations**

Evaluate the sum of the convergent infinite series: $24 - 12 + 6 - 3 + \cdots$

Step 1: Identify parameters: $a_1 = 24$, $r = \frac{-12}{24} = -\frac{1}{2}$. Since $|r| = \frac{1}{2} < 1$, the series converges. Step 2: Apply infinite sum formula: $S_\infty = \frac{a_1}{1-r} = \frac{24}{1-(-1/2)} = \frac{24}{3/2} = 24 \times \frac{2}{3} = \mathbf{16}$.

17 **[Challenging]** • **Target: Arithmetic series (sum formula)** • **Cross-Lesson: 5.01 Trigonometry & Polar Coordinates**

Evaluate the finite geometric sum: $\sum_{k=1}^{8} 3 \cdot 2^{k-1}$

Step 1: Identify components: Initial term $a_1 = 3(2^0) = 3$, ratio $r = 2$, number of terms $n = 8$. Step 2: Apply finite geometric sum formula: $S_8 = \frac{a_1(r^8 - 1)}{r - 1} = \frac{3(2^8 - 1)}{2 - 1} = 3(256 - 1) = 3(255) = \mathbf{765}$.

18 **[Challenging]** • **Target: Geometric sequences** • **Cross-Lesson: 6.01 Limits**

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

Use the infinite sum formula $S_\infty = \frac{a_1}{1-r}$: $25 = \frac{a_1}{1-(-0.4)} = \frac{a_1}{1.4} \Rightarrow a_1 = 25 \times 1.4 = 35$. Compute the 4th partial sum $S_4 = \frac{a_1(1-r^4)}{1-r}$: $(-0.4)^4 = 0.0256$. $S_4 = \frac{35(1-0.0256)}{1.4} = \frac{35(0.9744)}{1.4} = 25 \times 0.9744 = 24.36$.

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19 [Challenging] • Target: Geometric series (finite) • Cross-Lesson: 1.01 Complex Numbers

An arithmetic sequence has first term $a_1 = 4$ and common difference $d = 3$. If the sum of the first n terms is 175, find the value of n.

Step 1: Set up the sum formula: $S_n = \frac{n}{2}(2a_1 + (n-1)d) = 175$
$\frac{n}{2}(2(4) + (n-1)(3)) = 175 \Rightarrow n(8 + 3n - 3) = 350$ $n(3n + 5) = 350 \Rightarrow 3n^2 + 5n - 350 = 0.$ Step 2: Factor the quadratic equation: $(3n + 35)(n - 10) = 0$ Step 3: Reject non-integer solution: Since n must be a positive integer, $n = -35/3$ is rejected, leaving **$n = 10$**.

20 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 2.01 Matrices & Vectors

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.

Step 1: Determine first term and common ratio: First term $a_1 = 12$. Common ratio $r = \frac{8}{12} = \frac{2}{3}$. Step 2: Check convergence condition: Since $|r| = \left|\frac{2}{3}\right| = \frac{2}{3} < 1$, the infinite series converges. Step 3: Compute infinite sum: $S_\infty = \frac{a_1}{1-r} = \frac{12}{1-2/3} = \frac{12}{1/3} = 36$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arithmetic sequences

An arithmetic sequence (a_n) and a geometric sequence (b_n) of non-zero real numbers share the same first term, so $a_1 = b_1$. The sequences also satisfy $a_5 = b_2$ and $a_6 = b_3$, with $b_2 eq b_1$. If the maximum value of the partial sum $S_N = \sum_{k=1}^{N} a_k$ of the arithmetic sequence is 153, what is the sum of the infinite geometric series $\sum_{n=1}^{\infty} b_n$?

a. 64 ✓

b. 96

c. 144

d. 192

Let $a_1 = b_1 = a$, let d be the common difference of (a_n), and let r be the common ratio of (b_n). 1. Express the given terms: $a_5 = a + 4d = b_2 = ar$ $a_6 = a + 5d = b_3 = ar^2$ Subtracting the first equation from the second gives: $d = a_6 - a_5 = ar^2 - ar = ar(r-1)$ Substitute d back into $a + 4d = ar$:
$a + 4ar(r-1) = ar$ Since the terms of the sequence are non-zero, $aeq0$. Dividing by a:
$1 + 4r(r-1) = r \Rightarrow 4r^2 - 5r + 1 = 0$ $(4r-1)(r-1) = 0$ Since $b_2 eq b_1$.

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22 [Foundational] • Target: Arithmetic series (sum formula)

An infinite geometric sequence $a_1, a_2, a_3, \ldots$ of positive real numbers has common ratio $r < 1$ and sum S. The sub-sequence of terms whose indices are of the form $3k-1$ for positive integers k (that is, $a_2, a_5, a_8, \ldots$) has sum $\sum_{n=1}^{\infty} a_{3k-1} = \frac{3}{13}S$. Furthermore, the infinite series formed by the squares of the terms has sum $\sum_{n=1}^{\infty} a_n^2 = \frac{9}{2}S$. What is the value of $\sum_{n=1}^{\infty} na_n$?

a. $\frac{27}{2}$ ✓ **b.** $\frac{81}{4}$

c. $\frac{9}{2}$ **d.** $\frac{27}{4}$

Let the first term be $a_1 = a > 0$ and the common ratio be $r \in (0, 1)$. The sum of the original series is: $S = \sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$. **Step 1: Determine the common ratio r.** The sub-sequence consists of terms $a_2, a_5, a_8, \ldots$, which has first term $a_2 = ar$ and common ratio r^3. Its sum is: $\sum_{k=1}^{\infty} a_{3k-1} = \frac{ar}{1-r^3}$. Given that this sum is $\frac{3}{13}S$: $\frac{ar}{1-r^3} = \frac{3}{13} \cdot \frac{a}{1-r}$. Since $a > 0$ and.

23 [Foundational] • Target: Geometric sequences

An infinite geometric sequence of positive real numbers has first term a_1 and common ratio $r \in (0, 1)$. Let $S_1 = \sum_{n=1}^{\infty} a_n$, $S_2 = \sum_{n=1}^{\infty} a_n^2$, and $S_3 = \sum_{n=1}^{\infty} a_n^3$. If $S_1 = 4$ and $S_2 = 8$, what is the value of S_3?

a. $\frac{256}{13}$ ✓ **b.** $\frac{128}{13}$

c. $\frac{64}{7}$ **d.** 16

For an infinite geometric sequence with first term $a_1 > 0$ and ratio $r \in (0, 1)$, the terms are $a_n = a_1 r^{n-1}$. 1. The sum of the original sequence is: $S_1 = \frac{a_1}{1-r} = 4 \Rightarrow a_1 = 4(1-r)$ 2. The terms of the squared sequence are $a_n^2 = a_1^2 (r^2)^{n-1}$, which forms a geometric sequence with first term a_1^2 and common ratio r^2. Its sum is: $S_2 = \frac{a_1^2}{1-r^2} = 8$ 3. Relate S_2 and S_1 using the algebraic identity $1 - r^2 = (1-r)(1+r)$.

24 [Foundational] • Target: Geometric series (finite)

The second term of a geometric sequence of positive real numbers is k, and the fifth term is $k^2 - 20$. If the common ratio of the sequence is 2, what is the sum of the first 6 terms of the sequence?

a. 155 **b.** 160

c. **315** ✓ **d.** 630

Step 1: Express the terms in terms of the first term a_1 and common ratio $r = 2$. The n-th term of a geometric sequence is given by $a_n = a_1 r^{n-1}$. For the second term: $a_2 = a_1 (2)^{2-1} = 2a_1 = k \Rightarrow a_1 = \frac{k}{2}$ For the fifth term: $a_5 = a_1 (2)^{5-1} = 16a_1$ Substituting $a_1 = \frac{k}{2}$ gives: $a_5 = 16\left(\frac{k}{2}\right) = 8k$ Step 2: Solve for the parameter k. We are given $a_5 = k^2 - 20$, so: $k^2 - 20 = 8k \Rightarrow k^2 - 8k - 20 = 0$ Factoring the quadratic equation.

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25 [Foundational] • Target: Infinite geometric series

Does the infinite geometric series with $a_1 = 5$ and $r = -2$ converge?

a. No — $|r| \geq 1$, it diverges. ✓

b. Yes — it converges to $\frac{a_1}{1-r}$.

c. Yes — every geometric series converges.

d. Cannot be determined.

$|r| = 2 \geq 1$, so by Theorem 3.2 the series diverges (has no finite sum).

26 [Foundational] • Target: Arithmetic sequences

Find the sum of the first 8 terms of the arithmetic sequence with $a_1 = -2$ and $d = -2$.

a. -72 ✓

b. -64

c. -128

d. -70

$S_8 = \frac{8}{2}(2(-2) + (7)(-2)) = -72$.

27 [Foundational] • Target: Arithmetic series (sum formula)

The first three terms of a geometric sequence of positive real numbers are $k - 1$, $k + 2$, and $3k$, in that order. What is the value of the 6th term of this sequence?

a. 24

b. 48

c. 96 ✓

d. 192

Step 1: Set up the ratio property for consecutive terms of a geometric sequence. Since the sequence is geometric, the common ratio r satisfies: $r = \frac{a_2}{a_1} = \frac{a_3}{a_2} \Rightarrow \frac{k+2}{k-1} = \frac{3k}{k+2}$ Step 2: Solve the quadratic equation for k. Cross-multiplying gives: $(k + 2)^2 = 3k(k - 1)$ $k^2 + 4k + 4 = 3k^2 - 3k$ $2k^2 - 7k - 4 = 0$ Factoring the quadratic: $(2k + 1)(k - 4) = 0$ Thus, $k = -\frac{1}{2}$ or $k = 4$. Step 3: Discard the extraneous solution. The problem stat.

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28 [Foundational] • Target: Geometric sequences

In an arithmetic sequence, the 3rd term is $k + 7$, the 6th term is $3k - 1$, and the 11th term is $6k - 8$, where k is a real constant. What is the first term, a_1, of this sequence?

a. 6 ✓

b. 10

c. 19

d. 26

In an arithmetic sequence with first term a_1 and common difference d, the nth term is given by $a_n = a_1 + (n - 1)d$. Step 1: Express the common difference d in terms of k using the differences between the given terms: $a_6 - a_3 = 3d = (3k - 1) - (k + 7) = 2k - 8 \Rightarrow d = \frac{2k-8}{3}$ $a_{11} - a_6 = 5d = (6k - 8) - (3k - 1) = 3k - 7 \Rightarrow d = \frac{3k-7}{5}$ Step 2: Equate the two expressions for d to solve for k: $\frac{2k-8}{3} = \frac{3k-7}{5}$ $5(2k - 8) = 3(3k - 7)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Geometric series (finite)

In an arithmetic sequence, the first term is $a_1 = 4$ and the fifth term is $a_5 = 28$. What is the value of the tenth term, a_{10}?

a. 52

b. 56

c. **58** ✓

d. 64

The nth term of an arithmetic sequence is given by the formula: $a_n = a_1 + (n - 1)d$ where a_1 is the first term and d is the common difference. Using the fifth term $a_5 = 28$: $28 = 4 + (5 - 1)d \Rightarrow 28 = 4 + 4d \Rightarrow 4d = 24 \Rightarrow d = 6$ Now, calculate the tenth term a_{10}: $a_{10} = a_1 + (10 - 1)d = 4 + 9(6) = 4 + 54 = 58$ Common misconceptions: - Choosing 56 incorrectly assumes direct proportionality (e.g., $a_{10} = 2 \times a_5 = 2 \times 28$), which does not hold for arithmeti.

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30 [Intermediate] • Target: Infinite geometric series

The first four terms of an arithmetic sequence are 5, 12, 19, and 26. What is the 20th term of this sequence?

a. 131

b. **138** ✓

c. 140

d. 145

To find the 20th term of the arithmetic sequence, first identify the initial term and the common difference: 1. The first term is $a_1 = 5$. 2. The common difference d is found by subtracting consecutive terms: $d = 12 - 5 = 7$. The general formula for the nth term of an arithmetic sequence is: $a_n = a_1 + (n - 1)d$ Substitute $n = 20$, $a_1 = 5$, and $d = 7$: $a_{20} = 5 + (20 - 1)(7) = 5 + 19(7) = 5 + 133 = 138$ Common misconceptions: - Using n instead of $n - 1$ yields $5 + 20(7) = 145$. - Calculating.

31 [Intermediate] • Target: Arithmetic sequences

The first four terms of an arithmetic sequence are shown below. $4, 11, 18, 25, \ldots$ What is the 31st term of the sequence?

a. 210

b. **214** ✓

c. 217

d. 221

To find the 31st term of an arithmetic sequence, use the formula for the nth term: $a_n = a_1 + (n - 1)d$ 1. Identify the first term a_1: $a_1 = 4$ 2. Find the common difference d by subtracting consecutive terms: $d = 11 - 4 = 7$ 3. Substitute $n = 31$, $a_1 = 4$, and $d = 7$ into the formula: $a_{31} = 4 + (31 - 1)(7) = 4 + 30(7) = 4 + 210 = 214$ - A common error is using n instead of $n - 1$, which yields $4 + 31(7) = 221$. - Another common error is computing $31 \times 7 = 217$ or forgetting to add the.

32 [Intermediate] • Target: Arithmetic series (sum formula)

The bar chart below shows the values of the first four terms of an arithmetic sequence, $a_1, a_2, a_3,$ and a_4, where $a_1 = 6$, $a_2 = 11$, $a_3 = 16$, and $a_4 = 21$. What is the value of the 8th term, a_8, of this sequence?

a. 36

b. **41** ✓

c. 46

d. 48

1. Find the common difference d by subtracting consecutive terms: $d = a_2 - a_1 = 11 - 6 = 5$ Check consistency with subsequent terms: $16 - 11 = 5$ and $21 - 16 = 5$. 2. Use the standard formula for the n th term of an arithmetic sequence: $a_n = a_1 + (n - 1)d$ For the 8th term ($n = 8$): $a_8 = 6 + (8 - 1)(5) = 6 + 7(5) = 6 + 35 = 41$ Alternatively, continuing term-by-term: $a_5 = 26$, $a_6 = 31$, $a_7 = 36$, $a_8 = 41$.

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33 [Intermediate] • Target: Geometric sequences

A geometric sequence has $a_1 = 3$ and common ratio $r = 3$. Find a_3.

a. **27** ✓

b. 81

c. 30

d. -27

Applying Sequences & Series principles, $a_3 = a_1 r^{n-1} = 3(3)^2 = 27$. Therefore, the verified correct option is 27.

34 [Intermediate] • Target: Geometric series (finite)

An arithmetic sequence has $a_3 = 4$ and $a_7 = -4$. Find a_1.

a. **8** ✓

b. 6

c. -8

d. 7

$d = \frac{a_7 - a_3}{7 - 3} = \frac{-4 - 4}{4} = -2$. Then $a_1 = a_3 - (3 - 1)d = 4 - (2)(-2) = 8$.

35 [Intermediate] • Target: Infinite geometric series

An arithmetic series has $a_1 = 5$, $d = 3$, and $S_n = 390$. Find n.

a. **15** ✓

b. 16

c. 13

d. 30

Solve $390 = \frac{n}{2}(2(5) + (n - 1)(3))$ for n: $n = 15$ (the positive integer root).

36 [Intermediate] • Target: Arithmetic sequences

Find the common difference of the arithmetic sequence $-2,\ -4,\ -6,\ -8,\ \ldots$

a. **-2** ✓

b. 2

c. -1

d. -4

Subtract consecutive terms: $-4 - (-2) = -2$.

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37 [Intermediate] • Target: Arithmetic series (sum formula)

Find the common ratio of the geometric sequence $2,\ -6,\ 18,\ -54,\ \ldots$

a. **-3 ✓**

b. 3

c. -2

d. -8

Divide consecutive terms: $-6 \div 2 = -3$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Geometric sequences • Cross-Lesson: 6.01 Limits

Consider a geometric sequence where the first term is $a_1 = \frac{1}{2}$ and the common ratio is r. If the sum of the first three terms is $S_3 = \frac{7}{2}$, find the value of the common ratio r.

a. **2 ✓**

b. -2

c. 1/2

d. 3

The sum of the first three terms of a geometric sequence is given by the formula $S_n = a_1\frac{1-r^n}{1-r}$ for $req1$. We are given $a_1 = \frac{1}{2}$ and $S_3 = \frac{7}{2}$. Substituting these values into the formula: $\frac{7}{2} = \frac{1}{2} \cdot \frac{1-r^3}{1-r}$ Multiply both sides by 2: $7 = \frac{1-r^3}{1-r}$ We know that $1 - r^3 = (1 - r)(1 + r + r^2)$. So, for $req1$, we can simplify the expression: $7 = 1 + r + r^2$ Rearrange into a quadratic equation: $r^2 + r + 1 - 7 = 0$ $r^2 + r - 6 = 0$ Factor the.

39 [Challenging] • Target: Geometric series (finite) • Cross-Lesson: 1.01 Complex Numbers

A geometric sequence has its first term a_1 and a common ratio r. The sum of the first three terms is 26, and the sum of the first six terms is 728. If $a_1 = 2$, find the value of r.

a. 2

b. **3 ✓**

c. -2

d. -3

Let the first term of the geometric sequence be a_1 and the common ratio be r. The sum of the first n terms of a geometric sequence is given by $S_n = a_1\frac{1-r^n}{1-r}$ (for $req1$). We are given that $a_1 = 2$. The sum of the first three terms is $S_3 = 26$. So, $2\frac{1-r^3}{1-r} = 26$. This simplifies to $\frac{1-r^3}{1-r} = 13$. Using the difference of cubes formula, $1 - r^3 = (1 - r)(1 + r + r^2)$. Thus, $1 + r + r^2 = 13$. Rearranging this, we get a quadratic equation.

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40 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 2.01 Matrices & Vectors

Consider the arithmetic sequence with first term $a_1 = 5$ and a common difference $d = 3$. What is the formula for the n-th term, a_n?

a. $a_n = 5 + 3(n - 1)$ ✓

b. $a_n = 3n + 2$

c. $a_n = 5n + 3$

d. $a_n = 5 + 3n$

The formula for the n-th term of an arithmetic sequence is given by $a_n = a_1 + d(n - 1)$, where a_1 is the first term and d is the common difference. In this case, $a_1 = 5$ and $d = 3$. Substituting these values into the formula gives $a_n = 5 + 3(n - 1)$. If we simplify this, we get $a_n = 5 + 3n - 3 = 3n + 2$. Therefore, the direct formula is $a_n = 5 + 3(n - 1)$ or its simplified form $a_n = 3n + 2$. Choice A directly uses the definition. Choice B is the simplified form. Let's examine all options: Choice A.

41 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 4.01 Advanced Functions & Equations

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

a. 4

b. **5** ✓

c. 6

d. 7

The sum of the first n terms of a geometric sequence is given by the formula $S_n = a_1\frac{r^n - 1}{r - 1}$, where a_1 is the first term and r is the common ratio. Given $a_1 = 5$ and $r = 2$, and $S_n = 315$, we can substitute these values into the formula: $315 = 5\frac{2^n - 1}{2 - 1}$ $315 = 5\frac{2^n - 1}{1}$ $315 = 5(2^n - 1)$ Divide both sides by 5: $\frac{315}{5} = 2^n - 1$ $63 = 2^n - 1$ Add 1 to both sides: $63 + 1 = 2^n$ $64 = 2^n$ Since $2^6 = 64$, we have $n = 6$. Let's check the choices.

42 [Challenging] • Target: Arithmetic series (sum formula) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

An arithmetic sequence has $a_1 = 3$ and $d = 6$. If the sum of the first n terms is 300, find n.

a. 11

b. 9

c. 5

d. 10 ✓

$S_n = \frac{n}{2}(2a_1 + (n - 1)d) = 300$, and solving the quadratic gives $n = 10$.

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43 [Challenging] • Target: Geometric sequences • Cross-Lesson: 6.01 Limits

Find the sum of the first 5 terms of the arithmetic series with $a_1 = 1$ and $d = 4$.

a. **45** ✓

b. 49

c. 40

d. 50

$a_5 = 1 + (5-1)(4) = 17$. $S_5 = \frac{5}{2}(a_1 + a_n) = \frac{5}{2}(1 + 17) = 45$.

44 [Challenging] • Target: Geometric series (finite) • Cross-Lesson: 1.01 Complex Numbers

Find the sum of the first 3 terms of the geometric series with $a_1 = -1$ and $r = 3$.

a. -13 ✓

b. -3

c. -12

d. -14

$S_n = \frac{a_1(1-r^n)}{1-r} = \frac{-1(1-(3)^3)}{1-(3)} = -13$.

45 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 2.01 Matrices & Vectors

If the limit $\lim_{x \to 3} \frac{x^2 + kx - 15}{x^2 - 4x + 3} = L$ exists and is a finite real number, where k and L are constants, what is the value of $k + L$?

a. 2

b. 4

c. **6** ✓

d. 8

First, evaluate the limit of the denominator as $x \to 3$: $3^2 - 4(3) + 3 = 9 - 12 + 3 = 0$ Since the denominator approaches 0 and the limit L is a finite real number, the numerator must also approach 0 as $x \to 3$ to produce an indeterminate form of $\frac{0}{0}$. Setting the numerator equal to 0 at $x = 3$: $3^2 + 3k - 15 = 0 \Rightarrow 9 + 3k - 15 = 0 \Rightarrow 3k - 6 = 0 \Rightarrow k = 2$ Next, substitute $k = 2$ back into the rational function and factor both numerator and denomin.

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